

ECOS3991

Interdisciplinary Impact in Economics A **Economics of Conflict**

Week 11 - Costs and consequences of conflict

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Recap

- Week 1 – Introduction to the Course + What is conflict?
- Week 2 – What is economic development? Why do people fight?
- Week 3 – Why do people fight? (Continued)
- Week 4 – Causes of conflict (Part I)
- Week 5 – Causes of conflict (Part II)
- Week 7 – Causes of conflict (Part III)
- Week 8 – Costs and consequences of conflict (Part I)
- Week 9 – Costs and consequences of conflict (Part II)

Plan for today

In Week 9, we extended our discussion on consequences of conflict to social dimensions, and looked at consequences on the society as a whole.

Today we focus on how conflict affects individuals: their education, health, mental wellbeing, and economic preferences.

Blattman, C. and J. Annan (2010), "The consequences of child soldiering", *Review of Economics and Statistics* 92, 882–898.

- Consider the effect of military service...
- ...on human capital
- ...and labor market outcomes
- in Uganda.
- Their focus is especially on child soldiers.



Chris Blattman



Jeanie Annan

- Data: Primary data.
- Survey of War Affected Youth (SWAY) conducted during the war in northern Uganda
- Consisted of 741 males born between 1975 and 1991

- Identification strategy: based on forced abduction of youngsters to join the rebel army.
- The authors argue that the "who, what and where" of such forced abductions was very much random!
- Those forcibly abducted to fight (i.e. combatants) - the treated group
- Develop a comparison group of non-abducted youth in the war zone who, for exogenous reasons, were not abducted. (i.e. noncombatants)
- The control group is from the same birth year and birthplace as the treated group.
- Compare the differences in outcomes for these two groups.

- Identifying assumption: Those who were forcibly abducted were not different from those who were not forcibly abducted. (substantiated by a balance test)
- Note: Use discussions with rebel leaders and survey data to confirm
 - No evidence of youth voluntarily joining the rebels.
 - Randomness of forced abductions

Econometric specification

$$Y_i = \beta_0 + \tau \cdot T_i + X_i^S \cdot \beta_1 + \varepsilon_i,$$

- Y is the outcome variable.
- T is the indicator for treatment, which takes a value of 1 if the individual was abducted, and 0 if not.
- X is a set of control variables.
- What is the coefficient of interest?
- Can this be causally interpreted?

TABLE 3.—ESTIMATES OF THE AVERAGE TREATMENT EFFECT OF ABDUCTION

Dependent Variable	(1) ATE	(2) Nonabducted mean	(3) %Δ
<i>Educational and labor market outcomes</i>			
Years of education	−0.75 [0.17]***	7.6	−10%
Indicator for functional literacy	−0.15 [0.04]***	0.80	−19%
Indicator for any employment in the past month	0.03 [0.04]	0.61	5%
Indicator for capital- or skill-intensive work	−0.05 [0.02]**	0.12	−43%
Log (Daily wage)	−0.33 [0.15]**	n.a	n.a
<i>Psychosocial and health outcomes</i>			
Index of psychological distress	0.57 [0.20]***	3.8	15%
Indicator for top quartile of distress	0.11 [0.04]***	0.23	49%
Index of social support	−0.16 [0.14]	5.5	−3%
Indicator for hostile attitudes	0.03 [0.01]**	0.07	40%
Indicator for physical fights	−0.02 [0.02]	0.07	−29%

Each entry represents a separate WLS regression. All variables defined and described in table 1. *Significant at 10%. **Significant at 5%. ***Significant at 1%. Treatment is binary and equals 1 if ever abducted and 0 otherwise. The percentage change (%Δ) is calculated as the ATE relative to the mean value for nonabducted youth. Robust standard errors in brackets, clustered by sampling unit (location and abduction status). Controls in the WLS regressions include age and location dummies, age and location interactions, and pretreatment individual and household characteristics. Weighted by inverse sampling probability, inverse attrition probability, and inverse propensity score.

- They find widespread and persistent economic and educational impacts on child soldiers.
- Military service reduces schooling by nearly a year, skilled employment halves, and earnings drop by a third.
- This effect on education is largely due to time away from civilian schooling and work experience.
- This educational deficit also reduces labor market success: while abducted youth are just as likely to be employed, they are less likely to be engaged in skilled work and earn lower wages.
- Military service also leads to psychological distress among those exposed to severe war violence.

- Shemyakina (2011) *"The effect of armed conflict on accumulation of schooling: Results from Tajikistan"*
 - Studies the differences in regional and temporal exposure to the 1992–1998 armed conflict in Tajikistan..
 - to understand the effect of violent conflict on schooling outcomes.
 - Indicator of exposure to violence: Damage to a household's residences (from the 1999 Tajik Living Standards Survey)
 - Findings: Girls who were of school age during the conflict and lived in affected regions were less likely to complete their mandatory schooling than girls of the same age who lived in the regions relatively unaffected by conflict.
 - No effect on education of boys.



Olga Shemyakina

Table 5

Damage to household dwelling and determinants of school enrollment: by age group.

Variables and Controls	Boys			Girls		
	Ages 7–15	Ages 7–11	Ages 12–15	Ages 7–15	Ages 7–11	Ages 12–15
	(1)	(2)	(3)	(4)	(5)	(6)
Household dwelling damage	0.027 (0.035)	0.061 (0.040)	0.009 (0.044)	−0.119** (0.049)	−0.090** (0.038)	−0.141* (0.077)
HH owns > 0.1 hectare of land	0.019 (0.023)	0.025 (0.030)	0.018 (0.024)	0.025 (0.024)	−0.001 (0.023)	0.051 (0.036)
Distance to school	−0.011* (0.006)	−0.023** (0.012)	−0.005 (0.005)	−0.013 (0.008)	−0.016 (0.010)	−0.011 (0.014)
Parents' education (years)						
Mother	0.009** (0.005)	0.012** (0.006)	0.004 (0.007)	0.020*** (0.005)	0.012* (0.006)	0.026*** (0.007)
Father	0.010** (0.005)	0.007 (0.005)	0.015** (0.006)	0.020*** (0.005)	0.016** (0.006)	0.023*** (0.007)
Ln spline of household expenditure per capita (pce) (at median)						
≤ 12,842 rubles	0.126*** (0.037)	0.105** (0.043)	0.139*** (0.052)	0.023 (0.028)	0.038 (0.033)	0.004 (0.042)
> 12,842 rubles	−0.002 (0.002)	0.000 (0.003)	−0.003 (0.003)	0.007** (0.003)	0.003 (0.003)	0.012*** (0.004)
N adults ages 17–65						
Females	0.007 (0.009)	0.006 (0.012)	0.010 (0.012)	0.013 (0.011)	0.013 (0.012)	0.014 (0.017)
Males	−0.017** (0.008)	−0.024 (0.015)	−0.006 (0.009)	−0.016 (0.012)	−0.018 (0.015)	−0.018 (0.015)
Rural	0.034 (0.046)	0.000 (0.057)	0.076 (0.050)	0.028 (0.044)	0.010 (0.041)	0.04 (0.061)
Constant	−0.593* (0.349)	−0.348 (0.428)	−0.644 (0.437)	0.086 (0.259)	0.134 (0.315)	0.195 (0.403)
N	1580	834	746	1626	840	786
R-squared	0.079	0.094	0.071	0.112	0.071	0.136

Notes: Columns represent OLS coefficients. Robust standard errors are in parenthesis. Primary education—ages 7–11; general education—ages 12–15. All regressions contain community fixed effects at the raion (district) level, individual age dummies and controls for missing information on parents. Data source: TLSS 1999.

* Significant at 10%.

** Significant at 5%.

*** Significant at 1%.

- Akresh, Lucchetti and Thirumurthy (2012) "*Wars and child health: Evidence from the Eritrean–Ethiopian conflict*"
 - Use household survey data from Eritrea to estimate the effect of exposure to the 1998–2000 Eritrea–Ethiopia war on children's health.
 - War-exposed children have lower height-for-age Z-scores, which is a key indicator for negative health outcomes. This is observed for both boys and girls who are born during the war.



Richard Akresh



Leonardo Lucchetti



Harsha Thirumurthy

Table 2

Measuring the impact of war exposure on children's height-for-age Z-score.

Data source: 2002 Eritrea Demographic and Health Survey.

Dependent variable: height-for-age Z-score	[1]	[2]	[3]	[4]	[5]	[6]
War Region * Alive During War	−0.454*** [0.106]	−0.447*** [0.103]				
War Region * Born During War			−0.425*** [0.112]	−0.421*** [0.110]		
War Region * Born Before War			−0.334* [0.192]	−0.340* [0.188]		
Months of war exposure					−0.038*** [0.010]	−0.036*** [0.010]
Child is male	−0.023 [0.037]	−0.006 [0.036]	−0.023 [0.037]	−0.007 [0.036]	−0.017 [0.037]	0.000 [0.036]
Mother's height (cm)		0.046*** [0.004]		0.046*** [0.004]		0.046*** [0.004]
Head of household years of school		0.046*** [0.007]		0.046*** [0.007]		0.046*** [0.007]
Head of household is male		−0.073 [0.052]		−0.073 [0.052]		−0.076 [0.053]
Head of household age		0.003 [0.002]		0.003 [0.002]		0.003 [0.002]
Urban		0.197*** [0.073]		0.197*** [0.073]		0.199*** [0.073]
P-value testing equality between born during and born before war			0.458	0.503		
Child age fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Region-specific time trends	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5199	5199	5199	5199	5199	5199

Couttenier, M., Petrencu, V., Rohner, D. and M. Theonig (2019), "The Violent Legacy of Conflict: Evidence on Asylum Seekers, Crime, and Public Policy in Switzerland", *American Economic Review* 109, 4378-4425.

- Study the effect of past exposure to conflict on migrants' violent behaviours.
- "Crimes committed by migrants"
- Focus on asylum seekers in Switzerland.
- Extends beyond the direct impact of war: Examines the spillover effects of war in one country, on social aspects of another country.



Mathieu Couttenier



Veronica Petrencu



Dominic Rohner



Mathias Theonig

Research question:

- Does past exposure to conflict in origin countries makes migrants more violence-prone in their host country?

Empirical challenge: Selection

- Host country can attract more/less crime-prone individuals
- Individuals with more/less crime propensities might decide to migrate
- Pre-conflict characteristics of host country can impact crime after asylum seekers arrive

Identification strategy:

- Compare crime-propensities between groups (i.e. cohorts) of the same nationality and migration wave, with differing exposure to civil conflicts and mass killings (i.e. born before/after conflict in the origin country)
- Balance tests to confirm that selection into treatment was random

$$CP_{n,a,t} = \alpha \times kid[1-12]_{n,a,t} + \sum_{k=13}^{80+} \beta(k) \times expo(k)_{n,a,t} + \mathbf{FE}_{n,t} + \mathbf{FE}_a + \varepsilon_{n,a,t},$$

- Cohort level analysis
- $CP_{n,a,t}$: crime rate for a cohort of nationality n and age a , living in Switzerland in year t
- $Kid[1-12]_{n,a,t}$ identifies if the cohort was exposed to violence in origin country when they were below 12 years of age.
- $expo(k)_{n,a,t}$ identifies exposure to violence in the origin country, at ages beyond 12.

Key finding: Childhood exposure to conflict or mass killing significantly affect the future violent criminal behavior of men, but not of women

TABLE 5—VIOLENT CRIME PROPENSITY AND CONFLICT EXPOSURE

Sample:	Violent crime propensity			
	Full (1)	CC and MK (2)	CC (3)	MK (4)
<i>Panel A. Men</i>				
<i>kid</i> [1–12]	1.096 (0.461)	1.085 (0.454)		
<i>kid</i> [1–12] (<i>only CC</i>)			1.220 (0.478)	
<i>kid</i> [1–12] (<i>only MK</i>)				1.167 (0.591)
Age FE	Yes	Yes	Yes	Yes
Nationality $\times$ year FE	Yes	Yes	Yes	Yes
Observations	13,720	12,564	12,514	5,737
R^2	0.193	0.200	0.200	0.196
Sample mean (crime propensity)	3.09	3.09	3.11	2.59
	(5)	(6)	(7)	(8)
<i>Panel B. Women</i>				
<i>kid</i> [1–12]	0.026 (0.117)	0.033 (0.118)		
<i>kid</i> [1–12] (<i>only CC</i>)			–0.045 (0.129)	
<i>kid</i> [1–12] (<i>only MK</i>)				0.275 (0.203)
Age FE	Yes	Yes	Yes	Yes
Nationality $\times$ year FE	Yes	Yes	Yes	Yes
Observations	9,503	8,658	8,640	5,082
R^2	0.092	0.055	0.055	0.063
Sample mean (crime propensity)	0.30	0.27	0.27	0.32

Notes: OLS estimations based on the sample of adult cohorts of asylum seekers. Robust standard errors are clustered at the nationality level. The dependent variable stands for the violent crime propensity of a cohort of nationality (n) $\times$ age (a) $\times$ year (t). *kid* [1–12] is a binary measure of childhood exposure to civil conflict or mass killings (columns 1 to 2), to civil conflict only (column 3), or to mass killings only (column 4). In columns 2 to 4, the sample is restricted to cohorts originating from countries that have experienced civil conflict or mass killings since 1946. Panel A is confined to the male adult cohorts and panel B to the female adult cohorts of asylum seekers. All estimations include age fixed effects, nationality $\times$ year fixed effects and a set of binary variables coding for past exposure, but at later ages $k \in \{13, 14, 15, \dots, 80+\}$. Therefore, the reference group consists of those individuals born after the last years of violence. Abbreviations CC and MK refer respectively to civil conflict and mass killing.

- What we have discussed so far is just a drop in the ocean of the academic literature documenting the consequences of conflict.
- The key takeaway: War affects economic AND social aspects.
- Some of these are immediately visible, some only become visible years after.
- Most of these consequences are adverse. But some limited consequences of conflicts, such as increased political/community participation, are silver linings of an otherwise undesirable occurrence.

Format of the final exam

Format of the final exam

It is recommended that you familiarize yourself with the format of the exam, so you are well prepared ahead.

Mock final exam paper (no questions) available at Modules>Week 11

On the top right corner of the cover page, you are required to write your name, student number and the seat number.

These are extremely important. DO NOT SKIP.

You name will be digitally removed for anonymous marking.

Format of the final exam

You will have 10 minutes of reading time, and 1 hour of writing time.

This is a restricted open book exam: One A4 sheet of handwritten and/or typed notes, double-sided.

NO electronic devices are permitted.

There will be **2 questions**. Each question is of equal value. (15 marks per question)

Limit your answers to a maximum of 500 words per question.

You must write your answers in the spaces provided. Any writing outside of the spaces provided will not be graded.

There will be 3 blank pages at the end, for any additional writing.

If you are using these extra pages,, you **MUST** write “see additional page” in the relevant answer area, and clearly indicate the question number you are answering.

What will be tested?

Sample q

Policymaker X of country ABC states, "Climate has no impact on people's decision to engage in armed conflict." Evaluate this statement based on existing academic evidence on conflict, using at least 3 related academic articles.

The answer should contain the following:

- Refer to at least 3 academic articles that discuss the effect of climate on conflict?
- How do they causally identify the effect of climate on conflict?
- What do these articles find on the effect of climate on conflict?
- Based on this discussion, do you agree or disagree with this statement?

Wrapping up

- This is our last lecture. No lectures in Weeks 12 and 13.
- Week 11 tutorial assignment marks will be released very soon.
- Feedback tutorial in Week 12. No tutorials in Week 13.
- Media project is due on 29th May 23:59.
- Unit of Study Survey will open on 18th May (Monday of Week 12) and close on 7th June. You will receive an email notification when the survey opens. Please take a few minutes to record your feedback on the unit. The unit has massively benefitted from past students' feedback, and I truly appreciate your time and comments :)
- Please, also, feel free to email me with comments/questions/thoughts.

Wrapping up

Thank you for an enjoyable semester! I hope you enjoyed learning as much as I enjoyed teaching!

All the best!